Experiment B Analysis

This notebook reproduces the main statistical analysis for Experiment B.

Statistical Summary

Primary hypothesis testing methods:

Experiment A:

- Binomial test
- Confidence interval
- Cohen's h

Experiment B:

- Welch t-test
- Chi-square test
- Cramér's V
- Cohen's d

Files used

- `experiment_b_data_participant_summary.csv`
- `experiment_b_data_scenario.csv`

Main analyses

1. Group comparison: Matched vs Mismatched
2. Welch t-tests for participant-level outcomes
3. Effect sizes (Cohen's d)
4. 95% confidence intervals
5. Binary contact-again analysis
6. Persona-level summary
7. Scenario-level summary
8. Visualization export

In [4]:

```
import pandas as pd
import numpy as np
from scipy import stats
import math
import matplotlib.pyplot as plt
from pathlib import Path
from statsmodels.stats.multitest import multipletests

pd.options.display.float_format = '{:.4f}'.format

base = Path(".")
```

In [23]:

```
part = pd.read_csv(base / "../data/experiment_b_data_participant_summary.csv", sep=";", decimal=",", encoding="utf-8-sig")
part.columns = [c.strip() for c in part.columns]
part.head()
```

Out[23]:

	Participant_ID	Group	Persona	Total_Scenarios	Mean_Ease	Mean_Helpfulness	Mean_Trust	Mean_Contact_Likert	Mean_Style_Match	Overall_Mean	Contact_Binary_Rate	Overall_Helpfulness_Global	Confidence_Global
0	P001	Matched	STJ	4	4.2500	4.7500	5.0000	5.0000	5.0000	4.8000	1.0000	5	5
1	P002	Matched	STJ	4	3.7500	3.5000	3.7500	3.0000	3.0000	3.4000	0.7500	4	4
2	P003	Matched	STJ	4	4.2500	4.0000	4.0000	4.0000	4.0000	4.0500	0.7500	4	4
3	P004	Matched	STJ	4	3.7500	3.5000	3.5000	3.2500	3.2500	3.4500	0.5000	4	4
4	P005	Matched	STJ	4	5.0000	5.0000	5.0000	5.0000	4.7500	4.9500	1.0000	3	5

In [5]:

```
scen = pd.read_csv(base / "../data/experiment_b_data_scenario.csv", sep=";", decimal=",", encoding="utf-8-sig")
scen.columns = [c.strip() for c in scen.columns]
scen.head()
```

Out[5]:

	Participant_ID	Group	Persona	Scenario_ID	Ease	Helpfulness	Trust	Contact_Likert	Style_Match	Contact_Binary	Overall_Score_Scenario
0	P001	Matched	STJ	S1_Suspicious_Email	5	5	5	5	5	1	5.0000
1	P001	Matched	STJ	S2_Course_Schedule	4	5	5	5	5	1	4.5000
2	P001	Matched	STJ	S3_WiFi_Access	4	4	5	5	5	1	4.5000
3	P001	Matched	STJ	S4_Thesis_Topic	4	5	5	5	5	1	4.5000
4	P002	Matched	STJ	S1_Suspicious_Email	4	4	4	3	3	1	3.5000

In [25]:

```
# Group means
group_means = part.groupby("Group").agg(
    n=("Participant_ID", "count"),
    Mean_Ease=("Mean_Ease", "mean"),
    Mean_Helpfulness=("Mean_Helpfulness", "mean"),
    Mean_Trust=("Mean_Trust", "mean"),
    Mean_Contact_Likert=("Mean_Contact_Likert", "mean"),
    Mean_Style_Match=("Mean_Style_Match", "mean"),
    Overall_Mean=("Overall_Mean", "mean"),
    Contact_Binary_Rate=("Contact_Binary_Rate", "mean"),
    Overall_Helpfulness_Global=("Overall_Helpfulness_Global", "mean"),
    Confidence_Global=("Confidence_Global", "mean"),
).reset_index()
group_means
```

Out[25]:

	Group	n	Mean_Ease	Mean_Helpfulness	Mean_Trust	Mean_Contact_Likert	Mean_Style_Match	Overall_Mean	Contact_Binary_Rate	Overall_Helpfulness_Global	Confidence_Global
0	Matched	43	4.5087	4.4680	4.4535	4.4041	4.2965	4.4262	0.8953	4.1860	4.2791
1	Mismatched	37	4.1081	4.0203	4.1351	3.9392	3.6419	3.9689	0.7770	4.0000	4.0000

In [26]:

```
# Group comparison tests
def cohens_d_ind(a, b):
    a = np.asarray(pd.Series(a).dropna(), dtype=float)
    b = np.asarray(pd.Series(b).dropna(), dtype=float)
    na, nb = len(a), len(b)
    sa2, sb2 = a.var(ddof=1), b.var(ddof=1)
    sp = math.sqrt(((na-1)*sa2 + (nb-1)*sb2) / (na+nb-2))
    return (a.mean() - b.mean()) / sp if sp != 0 else np.nan

ordered_groups = [g for g in ["Matched", "Mismatched"] if g in part["Group"].unique()]
g1, g2 = ordered_groups[0], ordered_groups[1]

metrics = [
    "Mean_Ease", "Mean_Helpfulness", "Mean_Trust", "Mean_Contact_Likert",
    "Mean_Style_Match", "Overall_Mean", "Contact_Binary_Rate",
    "Overall_Helpfulness_Global", "Confidence_Global"
]

rows = []
for m in metrics:
    a = part.loc[part["Group"] == g1, m]
    b = part.loc[part["Group"] == g2, m]
    t = stats.ttest_ind(a, b, equal_var=False, nan_policy="omit")
    d = cohens_d_ind(a, b)
    rows.append({
        "Metric": m,
        f"{g1}_Mean": a.mean(),
        f"{g2}_Mean": b.mean(),
        "Mean_Difference": a.mean() - b.mean(),
        "t_stat": t.statistic,
        "p_value": t.pvalue,
        "Cohens_d": d
    })
tests_df = pd.DataFrame(rows)
tests_df
```

Out[26]:

	Metric	Matched_Mean	Mismatched_Mean	Mean_Difference	t_stat	p_value	Cohens_d
0	Mean_Ease	4.5087	4.1081	0.4006	2.6036	0.0113	0.5921
1	Mean_Helpfulness	4.4680	4.0203	0.4478	3.1944	0.0020	0.7136
2	Mean_Trust	4.4535	4.1351	0.3184	2.1079	0.0383	0.4712
3	Mean_Contact_Likert	4.4041	3.9392	0.4649	2.7530	0.0074	0.6217
4	Mean_Style_Match	4.2965	3.6419	0.6546	3.4979	0.0008	0.7972
5	Overall_Mean	4.4262	3.9689	0.4572	3.1092	0.0026	0.6976
6	Contact_Binary_Rate	0.8953	0.7770	0.1183	2.1156	0.0388	0.4925
7	Overall_Helpfulness_Global	4.1860	4.0000	0.1860	1.0339	0.3048	0.2359
8	Confidence_Global	4.2791	4.0000	0.2791	1.5397	0.1285	0.3534

In [27]:

```
# Mann–Whitney U Test for Matched vs Mismatched group comparisons
mw_results = []

metrics = [
    "Mean_Ease",
    "Mean_Helpfulness",
    "Mean_Trust",
    "Mean_Contact_Likert",
    "Mean_Style_Match",
    "Overall_Mean",
    "Contact_Binary_Rate",
    "Overall_Helpfulness_Global",
    "Confidence_Global"
]

for metric in metrics:
    matched_vals = part.loc[part["Group"] == "Matched", metric].dropna()
    mismatched_vals = part.loc[part["Group"] == "Mismatched", metric].dropna()

    mw_stat, mw_p = stats.mannwhitneyu(
      matched_vals,
      mismatched_vals,
      alternative="greater"
    )

    mw_results.append({
        "Metric": metric,
        "MW_Statistic": mw_stat,
        "MW_p_value": mw_p
    })

mw_df = pd.DataFrame(mw_results)

mw_df
```

Out[27]:

	Metric	MW_Statistic	MW_p_value
0	Mean_Ease	1042.0000	0.0070
1	Mean_Helpfulness	1106.5000	0.0011
2	Mean_Trust	1051.5000	0.0060
3	Mean_Contact_Likert	1080.5000	0.0026
4	Mean_Style_Match	1132.0000	0.0005
5	Overall_Mean	1114.5000	0.0010
6	Contact_Binary_Rate	953.0000	0.0398
7	Overall_Helpfulness_Global	868.5000	0.2215
8	Confidence_Global	923.5000	0.0866

In [6]:

```
# Binary contact-again analysis
contact_yes = scen.groupby("Group")["Contact_Binary"].agg(["sum", "count", "mean"]).reset_index()
contact_yes
```

Out[6]:

	Group	sum	count	mean
0	Matched	157	176	0.8920
1	Mismatched	115	148	0.7770

In [29]:

```
# Contingency table: Group × Contact_Binary
contact_table = pd.crosstab(scen["Group"], scen["Contact_Binary"])
contact_table
```

Out[29]:

Contact_Binary	0	1
Group		
Matched	19	157
Mismatched	33	115

In [30]:

```
# Chi-square test of independence
chi2, p_value, dof, expected = stats.chi2_contingency(contact_table)

chi2_df = pd.DataFrame({
    "Statistic": ["Chi2", "p_value", "df"],
    "Value": [chi2, p_value, dof]
})

chi2_df
```

Out[30]:

	Statistic	Value
0	Chi2	7.0631
1	p_value	0.0079
2	df	1.0000

In [31]:

```
# Expected frequencies
expected_df = pd.DataFrame(
    expected,
    index=contact_table.index,
    columns=contact_table.columns
)

expected_df
```

Out[31]:

Contact_Binary	0	1
Group		
Matched	28.2469	147.7531
Mismatched	23.7531	124.2469

In [32]:

```
# Effect size for 2x2 chi-square: Phi coefficient
n_total = contact_table.to_numpy().sum()
phi = np.sqrt(chi2 / n_total)

phi_df = pd.DataFrame({
    "Statistic": ["Phi"],
    "Value": [phi]
})

phi_df
```

Out[32]:

	Statistic	Value
0	Phi	0.1476

In [33]:

```
# Cramér's V effect size
cramers_v = np.sqrt(
    chi2 / (n_total * (min(contact_table.shape) - 1))
)

cramers_v_df = pd.DataFrame({
    "Statistic": ["Cramers_V"],
    "Value": [cramers_v]
})

cramers_v_df
```

Out[33]:

	Statistic	Value
0	Cramers_V	0.1476

In [34]:

```
# Odds ratio for contact-again binary outcome
# Assumes rows = groups, columns = Contact_Binary (0/1)

table = contact_table.copy()

# Reindex explicitly for safety
row_order = ["Matched", "Mismatched"] if "Mismatched" in table.index else ["Matched", "Mismatched"]
col_order = [0, 1]

table = table.reindex(index=row_order, columns=col_order)

a = table.loc[row_order[0], 1]  # Matched, Yes
b = table.loc[row_order[0], 0]  # Matched, No
c = table.loc[row_order[1], 1]  # Mismatched, Yes
d = table.loc[row_order[1], 0]  # Mismatched, No

odds_ratio = (a * d) / (b * c)

or_df = pd.DataFrame({
    "Statistic": ["Odds_Ratio"],
    "Value": [odds_ratio]
})

or_df
```

Out[34]:

	Statistic	Value
0	Odds_Ratio	2.3712

In [35]:

```
# Persona-level summary
persona_summary = part.groupby(["Persona", "Group"]).agg(
    n=("Participant_ID", "count"),
    Overall_Mean=("Overall_Mean", "mean"),
    Contact_Binary_Rate=("Contact_Binary_Rate", "mean"),
    Mean_Style_Match=("Mean_Style_Match", "mean"),
).reset_index()
persona_summary
```

Out[35]:

	Persona	Group	n	Overall_Mean	Contact_Binary_Rate	Mean_Style_Match
0	NFJ	Matched	8	4.6500	0.8750	4.4688
1	NFJ	Mismatched	4	4.1500	0.8125	4.0625
2	NTP	Matched	7	4.3429	0.9286	4.2857
3	NTP	Mismatched	7	4.1071	0.7143	3.7143
4	SFP	Matched	9	4.6667	0.9444	4.5556
5	SFP	Mismatched	13	3.7500	0.7692	3.4615
6	STJ	Matched	19	4.2487	0.8684	4.1053
7	STJ	Mismatched	13	4.0577	0.8077	3.6538

In [36]:

```
# Scenario-level summary
scenario_summary = scen.groupby(["Scenario_ID", "Group"]).agg(
    n=("Participant_ID", "count"),
    Ease=("Ease", "mean"),
    Helpfulness=("Helpfulness", "mean"),
    Trust=("Trust", "mean"),
    Contact_Likert=("Contact_Likert", "mean"),
    Style_Match=("Style_Match", "mean"),
    Contact_Binary_Rate=("Contact_Binary", "mean"),
).reset_index()
scenario_summary["Overall_Score"] = scenario_summary[
    ["Ease", "Helpfulness", "Trust", "Contact_Likert", "Style_Match"]
].mean(axis=1)
scenario_summary
```

Out[36]:

	Scenario_ID	Group	n	Ease	Helpfulness	Trust	Contact_Likert	Style_Match	Contact_Binary_Rate	Overall_Score
0	S1_Suspicious_Email	Matched	44	4.7500	4.7273	4.6364	4.6818	4.6591	1.0000	4.6909
1	S1_Suspicious_Email	Mismatched	37	4.2703	4.4054	4.2432	4.2432	3.7838	0.8108	4.1892
2	S2_Course_Schedule	Matched	44	4.1136	4.0455	4.1364	4.0000	3.7273	0.6818	4.0045
3	S2_Course_Schedule	Mismatched	37	4.0811	3.9730	4.1892	3.8919	3.6757	0.7568	3.9622
4	S3_WiFi_Access	Matched	44	4.6364	4.5455	4.4773	4.4091	4.3636	0.9318	4.4864
5	S3_WiFi_Access	Mismatched	37	3.8649	3.4865	3.7568	3.4324	3.2432	0.6757	3.5568
6	S4_Thesis_Topic	Matched	44	4.4773	4.5000	4.5227	4.4545	4.4545	0.9545	4.4682
7	S4_Thesis_Topic	Mismatched	37	4.2162	4.2162	4.3514	4.1892	3.8649	0.8649	4.1676

In [45]:

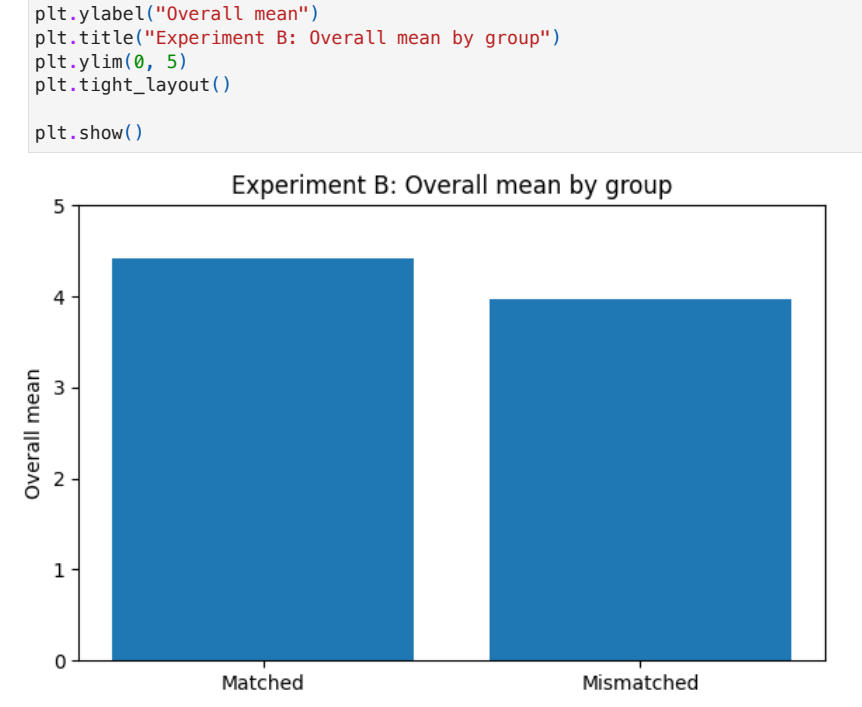
```
# Group comparison analysis (Matched vs Mismatched)
metrics = [
    "Mean_Ease",
    "Mean_Helpfulness",
    "Mean_Trust",
    "Mean_Contact_Likert",
    "Mean_Style_Match",
    "Overall_Mean",
    "Contact_Binary_Rate",
    "Overall_Helpfulness_Global",
    "Confidence_Global"
]

# Helper: Cohen's d for independent groups
def cohens_d_ind(a: pd.Series, b: pd.Series) -> float:
    a = np.asarray(a.dropna(), dtype=float)
    b = np.asarray(b.dropna(), dtype=float)

    na, nb = len(a), len(b)
    sa2 = a.var(ddof=1)
    sb2 = b.var(ddof=1)

    sp = math.sqrt(((na - 1) * sa2 + (nb - 1) * sb2) / (na + nb - 2))
    if sp == 0:
        return np.nan
    return (a.mean() - b.mean()) / sp

# Group names
g1 = "Matched"
g2 = "Mismatched"

rows = []

for metric in metrics:
    a = part.loc[part["Group"] == g1, metric]
    b = part.loc[part["Group"] == g2, metric]

    # Welch t-test
    t_res = stats.ttest_ind(a, b, equal_var=False, nan_policy="omit")

    mean_a = a.mean()
    mean_b = b.mean()
    mean_diff = mean_a - mean_b
    d = cohens_d_ind(a, b)

    # Welch-Satterthwaite confidence interval for mean difference
    na = a.dropna().shape[0]
    nb = b.dropna().shape[0]
    va = a.var(ddof=1)
    vb = b.var(ddof=1)

    se = math.sqrt((va / na) + (vb / nb))

    df = ((va / na) + (vb / nb)) ** 2 / (
        ((va / na) ** 2) / (na - 1) + ((vb / nb) ** 2) / (nb - 1)
    )

    t_crit = stats.t.ppf(0.975, df)
    ci_low = mean_diff - t_crit * se
    ci_high = mean_diff + t_crit * se

    rows.append({
        "Metric": metric,
        "Matched_Mean": mean_a,
        "Mismatched_Mean": mean_b,
        "Mean_Difference": mean_diff,
        "t_stat": t_res.statistic,
        "p_value": t_res.pvalue,
        "Cohens_d": d,
        "CI95_Low": ci_low,
        "CI95_High": ci_high
    })

group_comparison = pd.DataFrame(rows)

# Apply FDR correction to the original Welch t-test p-values
fdr_results = multipletests(
    group_comparison["p_value"],
    alpha=0.05,
    method="fdr_bh"
)

group_comparison["p_value_FDR"] = fdr_results[1]
group_comparison["Significant_FDR"] = fdr_results[0]

# Preview
group_comparison
```

Out[45]:

	Metric	Matched_Mean	Mismatched_Mean	Mean_Difference	t_stat	p_value	Cohens_d	CI95_Low	CI95_High	p_value_FDR	Significant_FDR
0	Mean_Ease	4.5087	4.1081	0.4006	2.6036	0.0113	0.5921	0.0937	0.7075	0.0203	True
1	Mean_Helpfulness	4.4680	4.0203	0.4478	3.1944	0.0020	0.7136	0.1687	0.7269	0.0079	True
2	Mean_Trust	4.4535	4.1351	0.3184	2.1079	0.0383	0.4712	0.0176	0.6191	0.0499	True
3	Mean_Contact_Likert	4.4041	3.9392	0.4649	2.7530	0.0074	0.6217	0.1284	0.8014	0.0167	True
4	Mean_Style_Match	4.2965	3.6419	0.6546	3.4979	0.0008	0.7972	0.2813	1.0280	0.0074	True
5	Overall_Mean	4.4262	3.9689	0.4572	3.1092	0.0026	0.6976	0.1644	0.7501	0.0079	True
6	Contact_Binary_Rate	0.8953	0.7770	0.1183	2.1156	0.0388	0.4925	0.0063	0.2304	0.0499	True
7	Overall_Helpfulness_Global	4.1860	4.0000	0.1860	1.0339	0.3048	0.2359	-0.1730	0.5451	0.3048	False
8	Confidence_Global	4.2791	4.0000	0.2791	1.5397	0.1285	0.3534	-0.0829	0.6411	0.1445	False

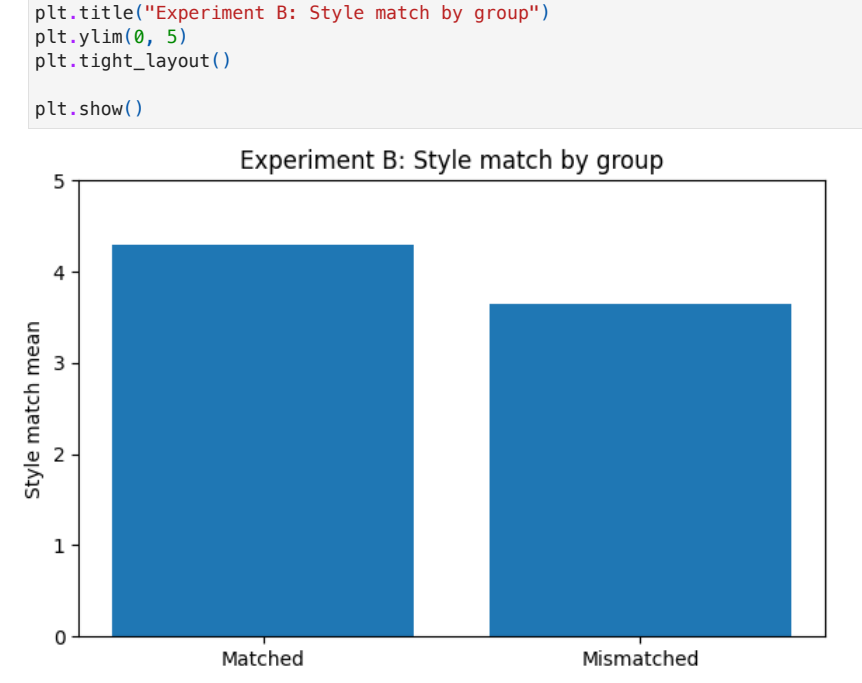
In [38]:

```
# Visualization 1: Group means
plt.figure(figsize=(6, 4))
plt.bar(group_means["Group"], group_means["Overall_Mean"])
plt.ylabel("Overall mean")
plt.title("Experiment B: Overall mean by group")
plt.ylim(0, 5)
plt.tight_layout()

plt.show()
```



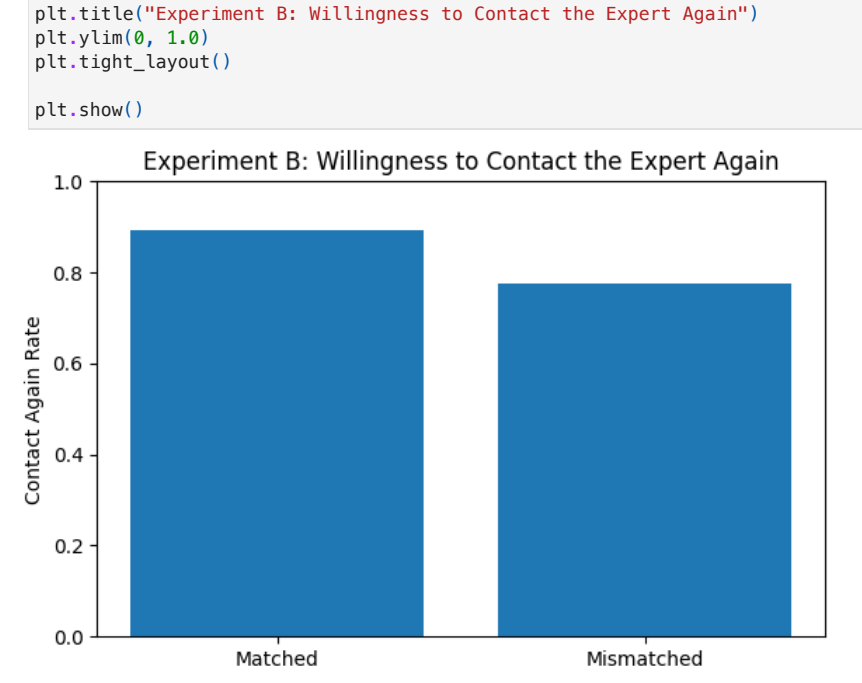
In [39]:

```
# Visualization 2: Style match by group
plt.figure(figsize=(6, 4))
plt.bar(group_means["Group"], group_means["Mean_Style_Match"])
plt.ylabel("Style match mean")
plt.title("Experiment B: Style match by group")
plt.ylim(0, 5)
plt.tight_layout()

plt.show()
```



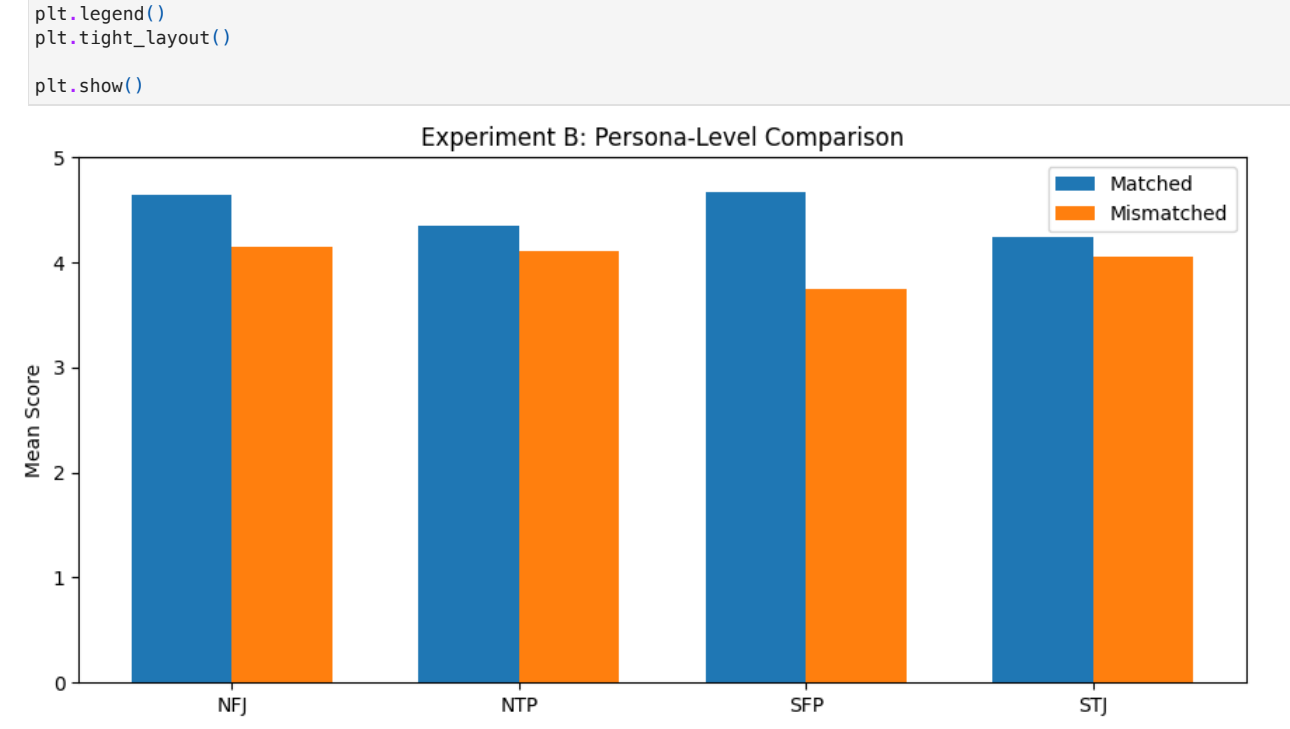
In [ ]:

```
# Visualization 3: Contact-again rate comparison
plt.figure(figsize=(6, 4))
plt.bar(contact_yes["Group"], contact_yes["mean"])
plt.ylabel("Contact Again Rate")
plt.title("Experiment B: Willingness to Contact the Expert Again")
plt.ylim(0, 1.0)
plt.tight_layout()

plt.show()
```



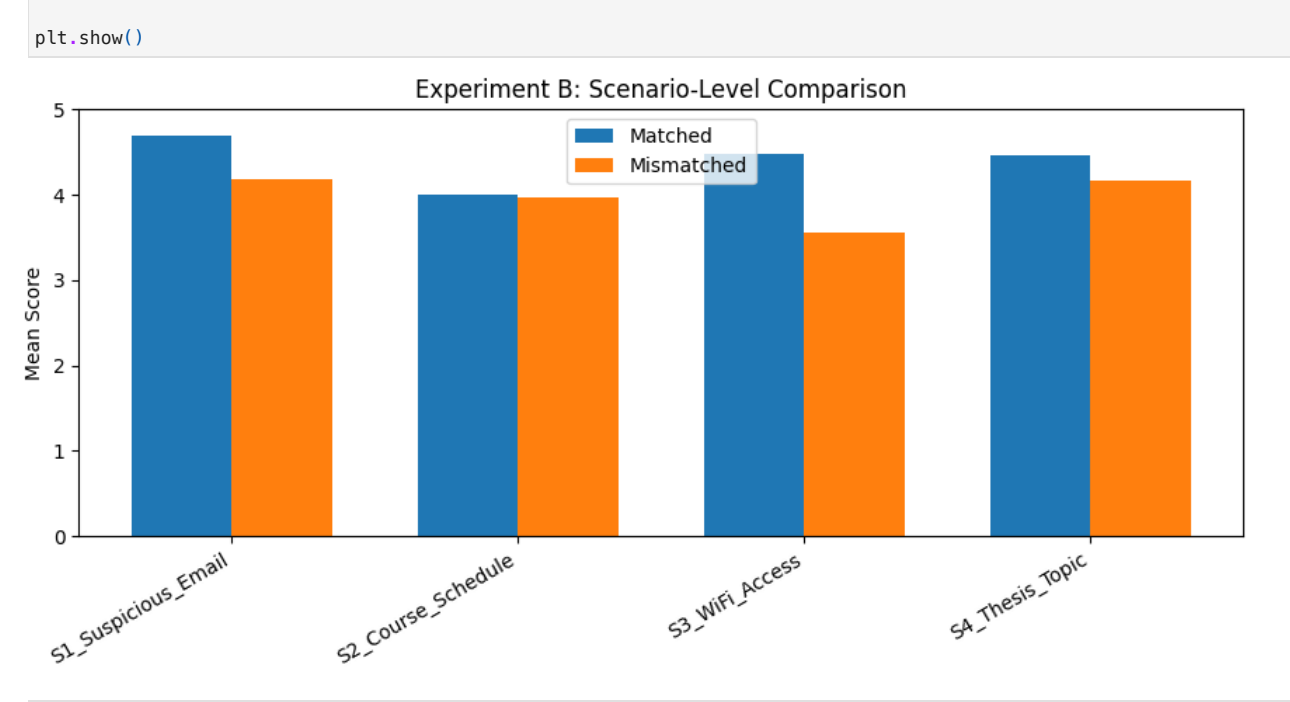
In [ ]:

```
# Visualization 4: Persona-level comparison
persona_pivot = persona_summary.pivot(index="Persona", columns="Group", values="Overall_Mean").fillna(np.nan)
x = np.arange(len(persona_pivot.index))
width = 0.35
plt.figure(figsize=(9, 4.5))
for i, col in enumerate(persona_pivot.columns[:2]):
    plt.bar(x + (i - 0.5) * width, persona_pivot[col], width=width, label=col)
plt.xticks(x, persona_pivot.index)
plt.ylabel("Mean Score")
plt.title("Experiment B: Persona-Level Comparison")
plt.ylim(0, 5)
plt.legend()
plt.tight_layout()

plt.show()
```



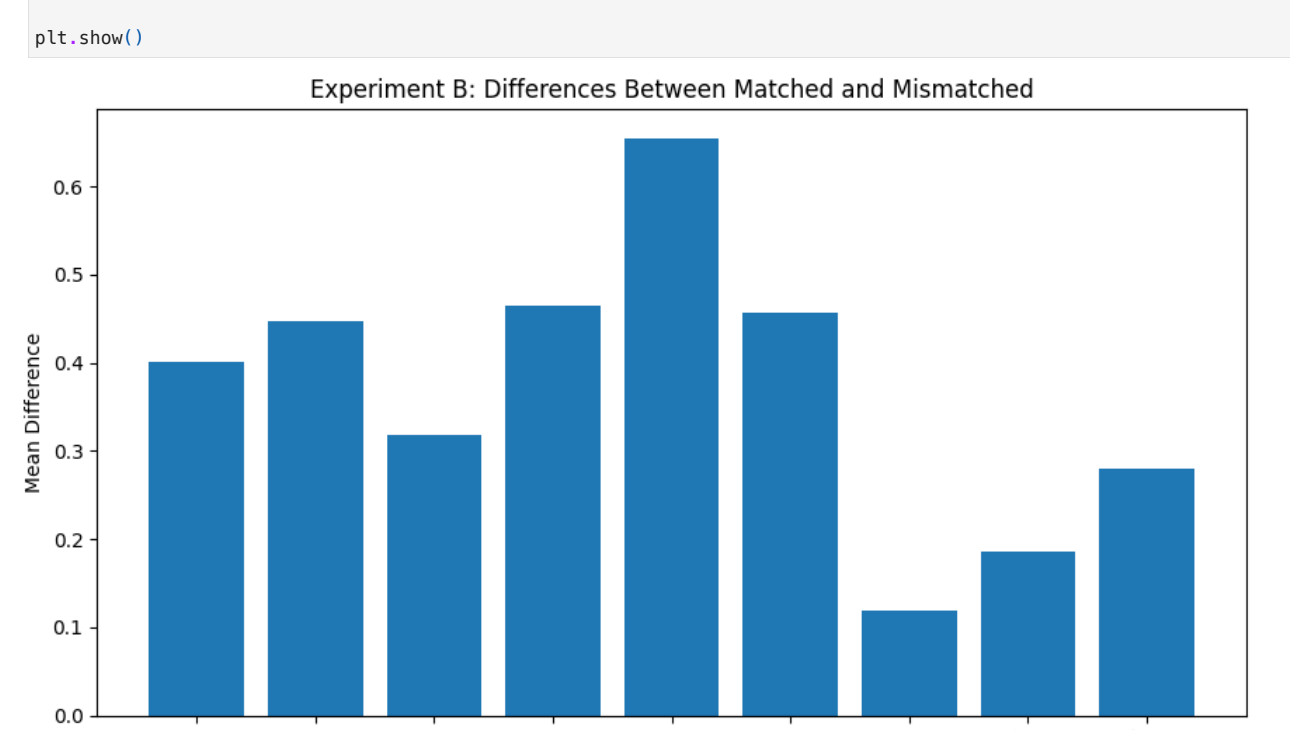
In [ ]:

```
# Visualization 5: Scenario-level comparison
scenario_pivot = scenario_summary.pivot(index="Scenario_ID", columns="Group", values="Overall_Score").fillna(np.nan)
x = np.arange(len(scenario_pivot.index))
width = 0.35
plt.figure(figsize=(9, 4.5))
for i, col in enumerate(scenario_pivot.columns[:2]):
    plt.bar(x + (i - 0.5) * width, scenario_pivot[col], width=width, label=col)
plt.xticks(x, scenario_pivot.index, rotation=30, ha="right")
plt.ylabel("Mean Score")
plt.title("Experiment B: Scenario-Level Comparison")
plt.ylim(0, 5)
plt.legend()
plt.tight_layout()

plt.show()
```

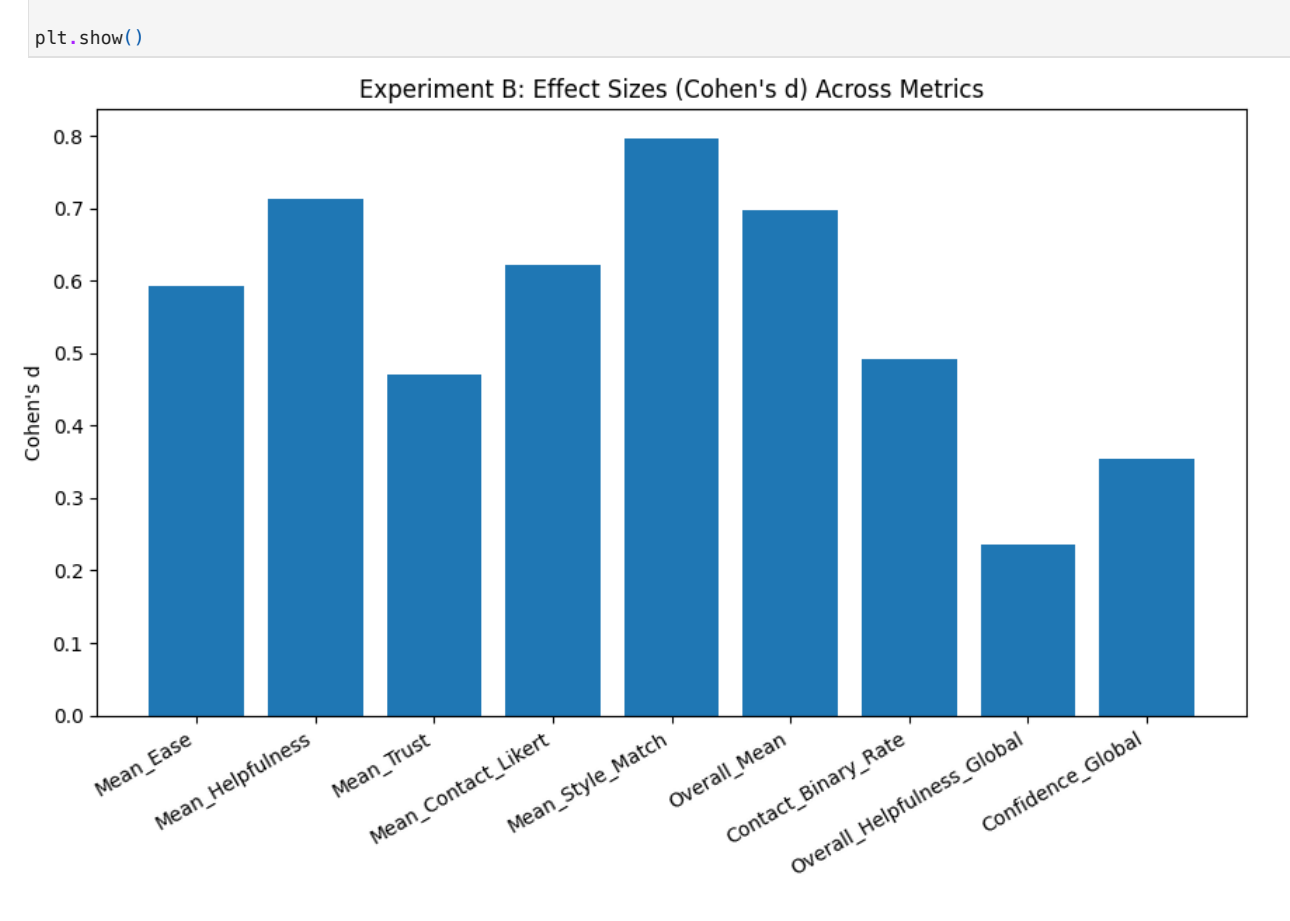


In [ ]:

```
# Visualization 6: Mean Differences Between Matched and Mismatched Conditions
x = np.arange(len(group_comparison.index))
plt.figure(figsize=(9, 6))
plt.bar(
  group_comparison["Metric"],
  group_comparison["Mean_Difference"]
)
plt.title("Experiment B: Differences Between Matched and Mismatched")
plt.ylabel("Mean Difference")
plt.xticks(x, group_comparison["Metric"], rotation=30, ha="right")
plt.xticks(rotation=30)
plt.tight_layout()

plt.show()
```



In [ ]:

```
# Visualization 7: Effect Sizes (Cohen's d) Across Metrics
x = np.arange(len(group_comparison.index))
plt.figure(figsize=(9, 6))
plt.bar(
  group_comparison["Metric"],
  group_comparison["Cohens_d"]
)
plt.title("Experiment B: Effect Sizes (Cohen's d) Across Metrics")
plt.ylabel("Cohen's d")
plt.xticks(x, group_comparison["Metric"], rotation=30, ha="right")
plt.xticks(rotation=30)
plt.tight_layout()

plt.show()
```